

ONLINE EXAMINATION PORTAL

A Web-Based Examination Management System

Project Documentation Report

1. Title of the Project

Online Examination Portal – A Web-Based Examination Management System with Automated Chapter-Balanced Question Paper Generation.

2. Abstract

The Online Examination Portal is a web-based application designed to make the examination process easier, faster, and more organized. It allows the admin or examiner to manage a chapter-wise question bank of around 200–400 questions and automatically generate a unique test for each student.

The main feature of the system is its question selection algorithm, which selects 20 questions from at least 5 different chapters. This ensures that every test has a balanced mix of questions and does not focus on only one topic. The portal also supports role-based access for the Main Admin, Teacher, and Student.

After the exam, the system evaluates the answers and generates performance analysis such as score, accuracy, and chapter-wise performance. However, students can view their score and analysis only if the Main Admin gives permission. This makes the system secure, fair, and useful for both examiners and students.

3. Introduction

Examinations are an essential component of every educational process as they evaluate the knowledge, understanding, and analytical ability of students. Traditional examination systems involve significant manual effort in setting question papers, distributing them, invigilating, evaluating answer scripts, and compiling results. These manual processes are time-consuming, prone to human error, and often delay the publication of results.

The widespread availability of internet access, modern web technologies, and reliable database systems has made it possible to design online examination platforms that automate every stage of the assessment cycle. An Online Examination Portal provides a centralized environment where examiners can prepare and curate questions, students can appear for tests from any device, and results are produced instantaneously.

The proposed Online Examination Portal goes a step beyond a typical quiz application. It introduces an automated paper-generation engine that intelligently selects twenty questions from at least five chapters of the question bank for every student. This guarantees content coverage, prevents leakage, and ensures fairness while reducing the workload on examiners.

4. Problem Statement

Traditional examination systems require manual paper setting, evaluation, and result preparation, which can be time-consuming and prone to errors. Many online quiz systems also select questions randomly without ensuring proper chapter-wise balance, which may result in an unfair or incomplete assessment.

Therefore, there is a need for a secure online examination portal that can automatically generate a unique 20-question paper from at least 5 different chapters, evaluate student responses instantly, and provide performance analysis under admin-controlled visibility.

5. Objectives of the System

- To design and develop a secure, web-based examination platform accessible through any standard browser.
- To maintain a centralized question bank of 200 to 400 questions arranged chapter-wise.
- To automatically generate a unique paper of twenty questions for each student, drawn from at least five different chapters.
- To support role-based access for Main Admin, Teacher, and Student with clearly defined permissions.
- To enable timed examinations with automatic submission on time-out.
- To evaluate objective answers instantly and store results in the database.
- To provide a detailed chapter-wise performance analysis to students, controlled by the admin.
- To enforce basic anti-cheating measures and secure session handling.

6. Scope of the Project

The Online Examination Portal is intended for colleges, universities, coaching institutes, and corporate training centers that conduct objective-type assessments. The scope of the current project includes the following capabilities:

- Centralized question bank management with chapter-wise organization.
- Automated, algorithmic generation of question papers for each candidate.
- Timed, browser-based examination with automatic answer saving.
- Instant evaluation of multiple-choice and true/false questions.
- Role-based dashboards for Main Admin, Teacher, and Student.
- Result generation with chapter-wise analysis and accuracy metrics.
- Permission-controlled visibility of scores and analysis.

The project does not currently include subjective answer evaluation, live webcam proctoring, or AI-based plagiarism detection. These are listed under future enhancements.

7. Existing System and Its Limitations

Existing examination practices, both offline and online, exhibit several limitations:

- Manual paper setting is time-consuming and dependent on the examiner's discretion.
- All candidates often receive the same question paper, enabling collusion and leakage.
- Random-selection quiz tools do not guarantee chapter diversity, leading to unbalanced tests.
- Manual evaluation delays result publication and increases the chance of errors.
- Result analysis is rarely chapter-wise, so students cannot identify weak areas.
- Most existing tools do not allow institutions to control whether students see their scores.
- Limited or no role separation between examiner, teacher, and student.

8. Proposed System

The proposed Online Examination Portal addresses the above limitations by introducing an automated, secure, and analytical examination environment. The Main Admin maintains a comprehensive question bank organized by chapters and configures exam rules such as the number of questions, eligible chapters, marks per question, time limit, and difficulty level. When a student begins an exam, the system invokes the Question Selection Algorithm, which dynamically constructs a unique paper of twenty questions drawn from at least five chapters.

The examination is conducted in the browser with a live countdown timer, automatic answer saving, and forced submission on time-out. As soon as the student submits, the system evaluates the responses, computes the total score, and produces a detailed chapter-wise breakdown. The student sees the score and analysis only if the Main Admin has granted score visibility; otherwise the result remains accessible only to authorized staff.

9. Key Features

- Chapter-wise question bank of 200–400 questions.
- Algorithmic paper generation with at least five chapters per paper.
- Three distinct user roles with granular permissions.
- Configurable exam rules: number of questions, marks, time, chapters, difficulty.
- Real-time timer with automatic submission.
- Instant evaluation and result storage.

- Chapter-wise performance analysis and weak-area identification.
- Admin-controlled visibility of scores and analysis to students.
- Secure JWT-based authentication and role-based access control.
- Basic anti-cheating measures such as full-screen mode and tab-switch detection.

10. User Roles and Permissions

10.1 Main Admin / Examiner

- Add, edit, and delete questions in the question bank.
- Organize questions chapter-wise and tag difficulty levels.
- Configure exam rules: number of questions, chapters, marks, time, difficulty mix.
- View complete performance data of every student.
- Grant or revoke permission for students to view scores and analysis.
- Manage teacher and student accounts.

10.2 Admin / Teacher

- Manage assigned subset of questions if permission is granted.
- Monitor ongoing exam sessions in real time.
- View performance reports of permitted student groups.

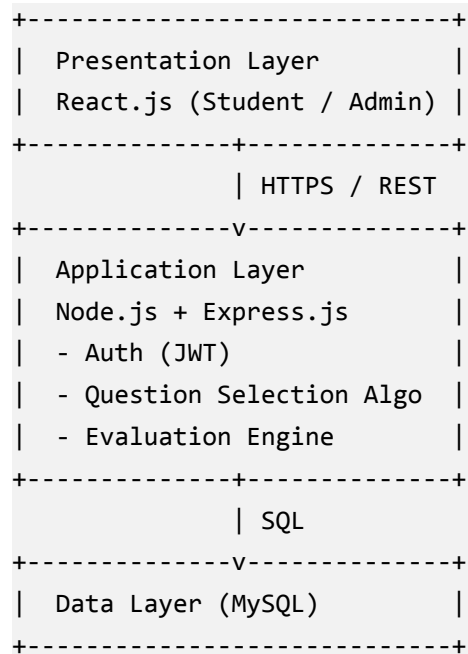
10.3 Student / Examinee

- Log in securely and view assigned exams.
- Attempt the algorithmically generated paper of twenty questions.
- Save answers and submit the test within the time limit.
- View score and analysis only if Main Admin has enabled visibility.

11. System Architecture

The portal follows a classic three-tier architecture comprising the Presentation Layer, the Application Layer, and the Data Layer. The Presentation Layer is implemented using React.js and communicates with the backend over HTTPS using RESTful JSON APIs. The Application Layer is built on Node.js with the Express.js framework and contains all business logic, the question-selection algorithm, evaluation logic, and authentication middleware. The Data Layer uses MySQL as the relational database to store users, roles, questions, exams, attempts, and results.

Architecture Diagram



12. Technology Stack

Layer	Technology
Frontend	React.js, React Router, Axios, Tailwind CSS
Backend	Node.js with Express.js framework
Database	MySQL (relational)
Authentication	JSON Web Tokens (JWT) with bcrypt password hashing
API Style	RESTful JSON APIs
Version Control	Git and GitHub

13. Database Design

The database consists of the following primary tables. Foreign keys are used to maintain relational integrity between users, exams, questions, and results.

13.1 Users

Column	Type	Description
user_id	INT (PK)	Unique user identifier
name	VARCHAR(100)	Full name

Column	Type	Description
email	VARCHAR(150)	Login email (unique)
password_hash	VARCHAR(255)	Bcrypt-hashed password
role_id	INT (FK)	References Roles
created_at	DATETIME	Account creation time

13.2 Roles

Column	Type	Description
role_id	INT (PK)	Role identifier
role_name	VARCHAR(50)	MAIN_ADMIN, TEACHER, STUDENT

13.3 Chapters

Column	Type	Description
chapter_id	INT (PK)	Chapter identifier
chapter_name	VARCHAR(150)	Title of the chapter
subject	VARCHAR(100)	Subject the chapter belongs to

13.4 QuestionBank

Column	Type	Description
question_id	INT (PK)	Question identifier
chapter_id	INT (FK)	References Chapters
question_text	TEXT	The question statement
option_a/b/c/d	VARCHAR(255)	Four options
correct_option	CHAR(1)	A, B, C or D
difficulty	ENUM	EASY, MEDIUM, HARD
marks	INT	Marks allotted

13.5 Exams

Column	Type	Description
exam_id	INT (PK)	Exam identifier
title	VARCHAR(150)	Exam title
total_questions	INT	Default 20

Column	Type	Description
min_chapters	INT	Default 5
duration_min	INT	Time limit
created_by	INT (FK)	Admin who created the exam

13.6 ExamQuestions

Stores the actual questions generated by the algorithm for each student attempt, ensuring traceability.

Column	Type	Description
exam_question_id	INT (PK)	Identifier
attempt_id	INT (FK)	References StudentAttempts
question_id	INT (FK)	References QuestionBank
question_order	INT	Position in the paper

13.7 StudentAttempts

Column	Type	Description
attempt_id	INT (PK)	Attempt identifier
student_id	INT (FK)	References Users
exam_id	INT (FK)	References Exams
started_at	DATETIME	Start time
submitted_at	DATETIME	Submission time
status	ENUM	IN_PROGRESS, SUBMITTED, EXPIRED

13.8 Results

Column	Type	Description
result_id	INT (PK)	Result identifier
attempt_id	INT (FK)	References StudentAttempts
total_score	INT	Marks obtained
correct_count	INT	Number of correct answers
incorrect_count	INT	Number of incorrect answers
accuracy	DECIMAL(5,2)	Percentage accuracy
chapter_analysis	JSON	Chapter-wise breakdown

13.9 Permissions

Column	Type	Description
permission_id	INT (PK)	Identifier
role_id	INT (FK)	References Roles
permission_key	VARCHAR(80)	e.g. VIEW_SCORE, EDIT_QUESTIONS
is_enabled	BOOLEAN	Permission state

14. Exam Flow

1. Student logs in using email and password; JWT is issued.
2. Student selects an assigned exam from the dashboard.
3. Backend invokes the Question Selection Algorithm and stores the generated paper.
4. Frontend loads the twenty questions one by one or in a paginated layout.
5. A countdown timer starts based on the configured duration.
6. Answers are auto-saved as the student selects options.
7. Student submits the test, or the system auto-submits on time-out.
8. Backend evaluates answers, computes score and chapter-wise analysis.
9. Result is stored in the database; visibility depends on admin permission.

15. Admin Flow

10. Main Admin logs in and accesses the admin dashboard.
11. Adds or imports questions into the chapter-wise question bank.
12. Creates a new exam, defining number of questions, chapters, marks, duration, and difficulty mix.
13. Monitors ongoing student attempts in real time.
14. Reviews complete performance reports of all students.
15. Toggles score-visibility permission for individual students or globally.

16. Student Performance Analysis

After evaluation, the system generates a comprehensive performance report containing the following metrics:

- Total score obtained and maximum possible score.

- Chapter-wise score and percentage.
- Number of correct and incorrect answers.
- Total time taken to complete the exam.
- Identified weak chapters (those with the lowest percentage).
- Overall accuracy percentage.

This analysis is shown to the student only when the Main Admin has enabled score visibility. Otherwise, the student sees a simple confirmation that the test has been submitted, while the full analysis remains available to admins and teachers.

17. Security Features

- Passwords stored as bcrypt hashes; never in plain text.
- JWT-based stateless authentication with short expiry and refresh tokens.
- Role-based access control enforced on every API endpoint.
- Students cannot access the question bank or other students' results.
- Exam sessions validated on every request to prevent tampering or replay.
- HTTPS enforced for all client-server communication.
- Basic anti-cheating: full-screen lock, tab-switch detection, copy/paste disabling, single-session enforcement.
- Audit logs of admin actions on question bank and permissions.

18. Advantages of the System

- Eliminates manual effort in paper setting and evaluation.
- Guarantees balanced coverage across multiple chapters.
- Generates unique question papers for every student, reducing malpractice.
- Instant result generation and detailed analytics.
- Centralized control of permissions and score visibility.
- Scalable to large numbers of concurrent users.
- Environmentally friendly and paperless.

19. Limitations

- Requires stable internet connectivity for all participants.
- Currently supports only objective-type questions (MCQ, True/False).

- Does not include live human or AI-based proctoring.
- Quality of the assessment depends on the quality of the question bank.
- Browser-based anti-cheating measures can be bypassed by determined users without hardware-level proctoring.

20. Future Enhancements

- AI-based question selection that adapts to student ability over time.
- Live proctoring with webcam and microphone monitoring.
- Plagiarism and cheating detection using behavioural analytics.
- Adaptive difficulty exams that adjust difficulty based on the student's performance.
- Automatic PDF report generation for results and analytics.
- Support for subjective questions with NLP-assisted evaluation.
- Mobile applications for Android and iOS.

21. Conclusion

The Online Examination Portal presents a complete, modern solution for conducting objective examinations in an automated, fair, and analytical manner. By combining a chapter-organized question bank with an intelligent paper-generation algorithm, the system ensures that every student receives a unique and topically balanced question paper. The role-based architecture gives institutions full control over the examination process, while detailed performance analytics empower students to identify and improve their weak areas—if the institution chooses to share that information.

With its React.js frontend, Node.js and Express.js backend, MySQL database, and JWT-based security, the portal is built on a robust and widely adopted technology stack. Although the present version focuses on objective assessments and basic anti-cheating measures, the modular architecture makes it straightforward to extend the platform with AI-based question selection, webcam proctoring, adaptive difficulty, and richer reporting in future iterations. The project therefore stands as a practical, scalable, and academically meaningful contribution to the field of online assessment.